The Islamia University of Bahawalpur

Software Requirements Specification

For

Toolkit with Games

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1. Introduction

1.1 Purpose

The main objective of this application is to have all in one kind of situation in which user doesn't has to use different resources to use these types of tools. The main purpose of this application is that different commonly used tools will available for user so he/she can easily have an advantage of using multiple tools in one place.

Also, it contains different games for free time play or usually for kids that helps them in their educational development.

1.2 Scope

The basic objective of developing this type of application is to enhance the use of desktop application. We know that now a days mostly games are online, and trend of Desktop Application is low but in Many Underdeveloped Nations where the internet facility is not available. So, that is the purpose we can develop this Desktop Application for Kid's, and the tools for daily basis needs.

And also, the improvement will be made according to the user need and scope of this application.

i. User

✓ Provide accurate results.

✓ Allow users to interact easily.

✓ Allow users to use simple and fast tools.

✓ Allow users to play games with no complexity.

✓ Easy to manage their records.

Advantages

✓ Provide effective interface.

✓ The software is able to run on all windows-based devices and environments.

✓ Installation may or may not require.

- ✓ More efficient & reliable.
- ✓ No internet required.
- ✓ Less time consuming and easy to use & consumes less computer memory.
- ✓ Since it is desktop application, all its use is done on the device without any additional resources.
- ✓ It does not take up space on the device.

Disadvantages

- ✓ If the device contains harmful viruses, then normal working may be affected.

1.3 Definitions, Acronyms, and Abbreviations.7

Term	Description
GUI	Graphical User Interface
Interactive system	A system which allows user to manipulate settings.
C#	Programming Language.
SRS	Software Requirement Specification
Cryptography	Method of Encryption & Decryption
BMI	Body Mass Index

1.4 References

1) www.google.com

2) www.slideshare.com

1.5 Overview

The overview of our desktop application is that it is an offline interface where a user can have an interacted environment where he or she can use multiple tools which helps him or her in daily

basis need and also this desktop application has different offline games that can help kids in their mental and educational growth.

2. The Overall Description

2.1 Product Perspective

A perfect application for students which helps in their study and extra curriculum activities.

The major objective of developing this type of application is for kids. This application has maximum performance and a better graphics that can run on an average system. It's very useful for kid's for educational purpose and for fun. By using this application, the kids can learn basic knowledge of English.

The system's aim is to provide better interface for commonly used tools, games and customer satisfaction by providing the following services such as:

Tools: -

Paint App: A drawing tool in which user can easily make simple drawings.

Encryption of Textual Data: Main purpose of this feature is to encrypt the important data. User can easily enter the text and with one click he gets the encrypted version of the data.

Notepad Editor: A fully functional notepad application used for text editing.

Heartbeat Rate Calculator: A basic tool in which user can be able to find his/her minimum and maximum heartbeat target rate.

BMI (Body Mass Index): A tool through which user can be able to calculate his/her BMI values and description of low, high, or normal.

Calculator: A basic tool in which user can be able to perform calculation and get accurate results.

Binary Decimal Converters: A tool through which user can be able to perform different Number System conversions like:

- Binary to Decimal
- Decimal to Binary
- Decimal to Octal
- Decimal to Hexadecimal etc.

Unit Conversion: A tool through which user can be able to perform different units' conversion like:

- Celsius to Fahrenheit
- Fahrenheit to Celsius
- Kilograms to Pounds
- Kilometers to Meters etc.

And many more tools…

Games: -

Matching Picture Game: A game in which the user has to match same pair of images in order to complete the game.

Speed Break Game: A game in which the user has to stop the fastly moving images for three different times. Same match of images makes you win.

Word Guessing Game: A game in which the user has to guess the word with the help of some riddle.

And many more games…

2.2 Product Functions

1) High Performance
2) User Convenience
3) Security
4) Smooth Integration
5) Quality Assurance
6) Long Term Maintenance

2.3 User Characteristics

Guest User:

User can easily view game levels and change their settings.

2.4 General Constraints

- Accessibility: User easily open the application
- Memory constraints: laptop must have 512 mb Ram.
- Tools: Desktop Application by using C#, My SQL.

2.5 Assumptions and Dependencies

For developing this type of application, we need a laptop or PC. The application can run smoothly if you have either of these platforms like PC or Laptop.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 System Interfaces

Our system has single type of interface

1) User

3.1.2 Interfaces

1. User can view the game interface
2. Change Settings

3.1.3 Hardware Interfaces

1) Minimum Ram Required is 512 mb

2) Pentium 4 pc or above

3.1.4 Software Interfaces

1) Language C#
2) Windows 7,8,10

3.2 Functional Requirements

3.2.1 <Functional Requirement of view Register Users Accounts>

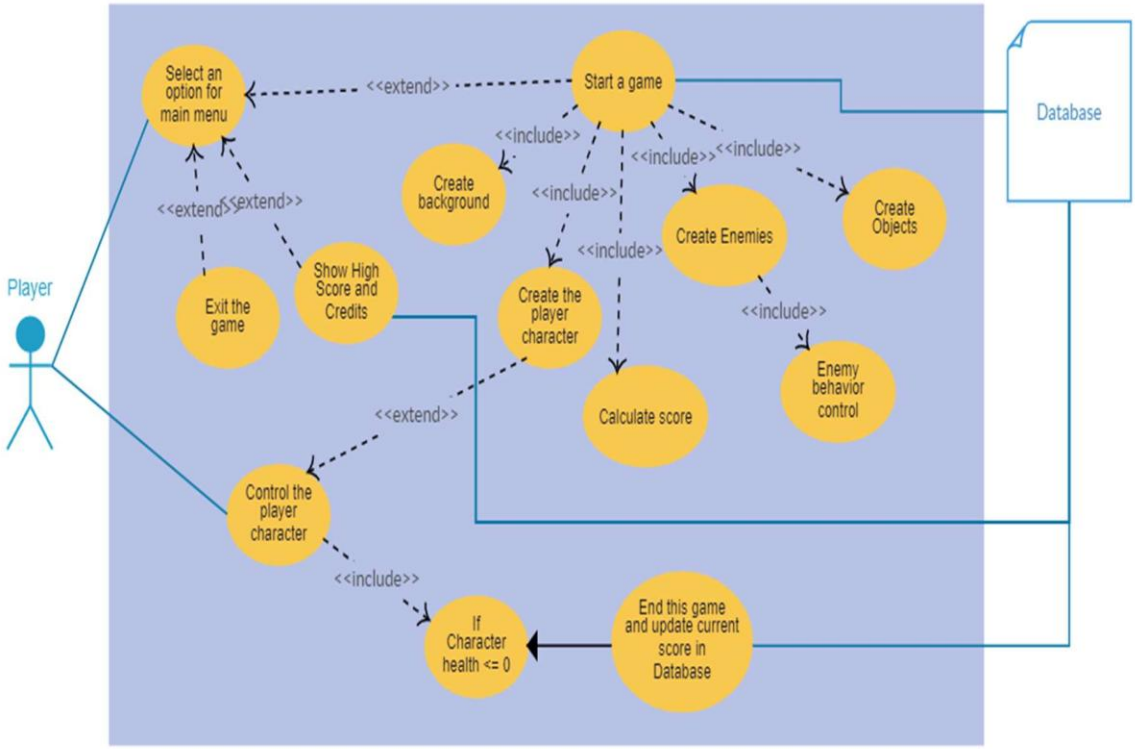
3.2.1.1 User can view all the information in application.

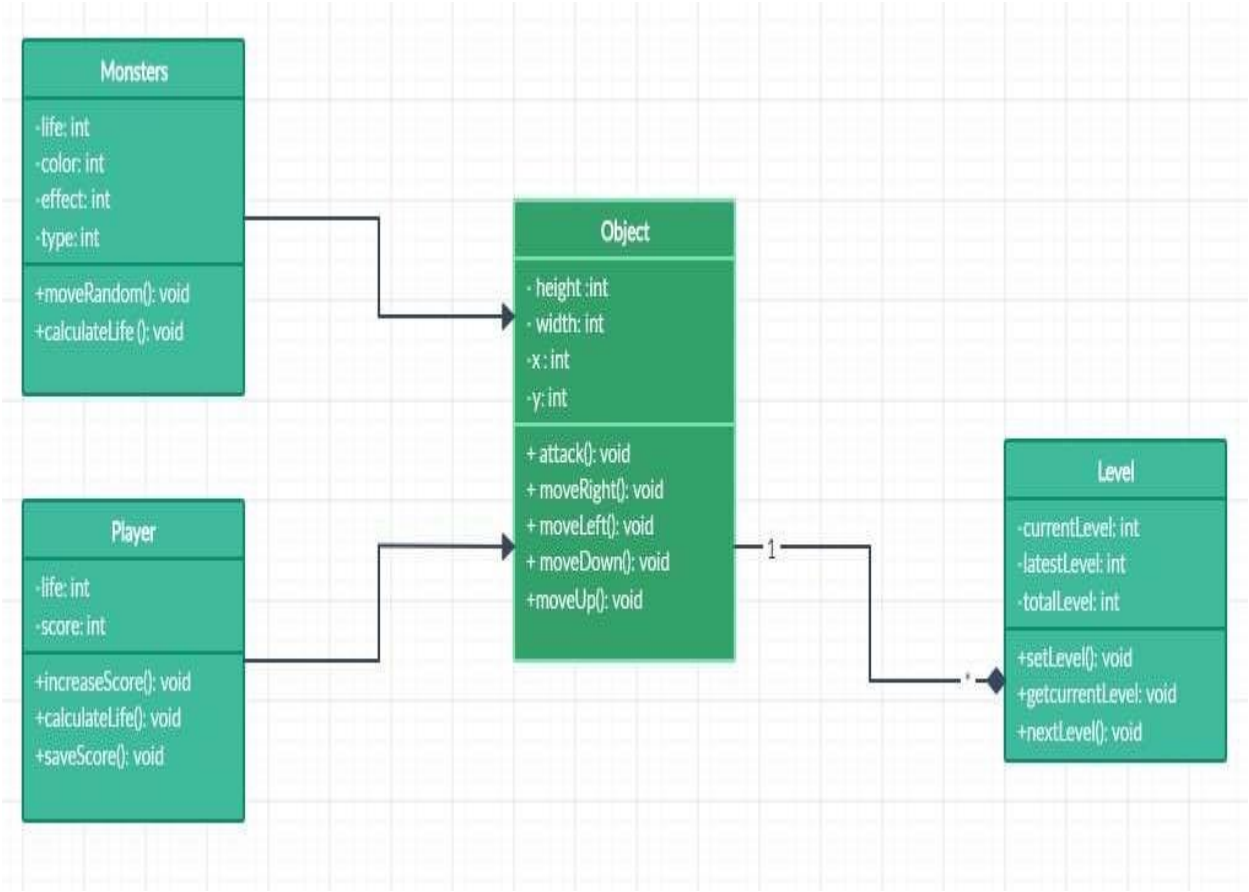
3.2.1.2 User can play different levels of game.

3.2.1.3 User can change their setting according their perspective.

3.2.1.4 User can get required result.

3.3 Use Cases


Select an option for main menu
<<extend>>
Start a game
Database
<<include>>
<<include>>
<<include>>
Create background
<<include>>
Create Objects
<<extend>>
<<extend>>
Create Enemies
<<include>>
Player
Show High Score and Credits
Exit the game
Create the player character
<<include>>
Enemy behavior control
Calculate score
<<extend>>
Control the player character
<<include>>
If Character health <= 0
End this game and update current score in Database



3.4 Classes / Objects

3.5 Non-Functional Requirements

3.5.1 Performance

3.5.2 Reliability

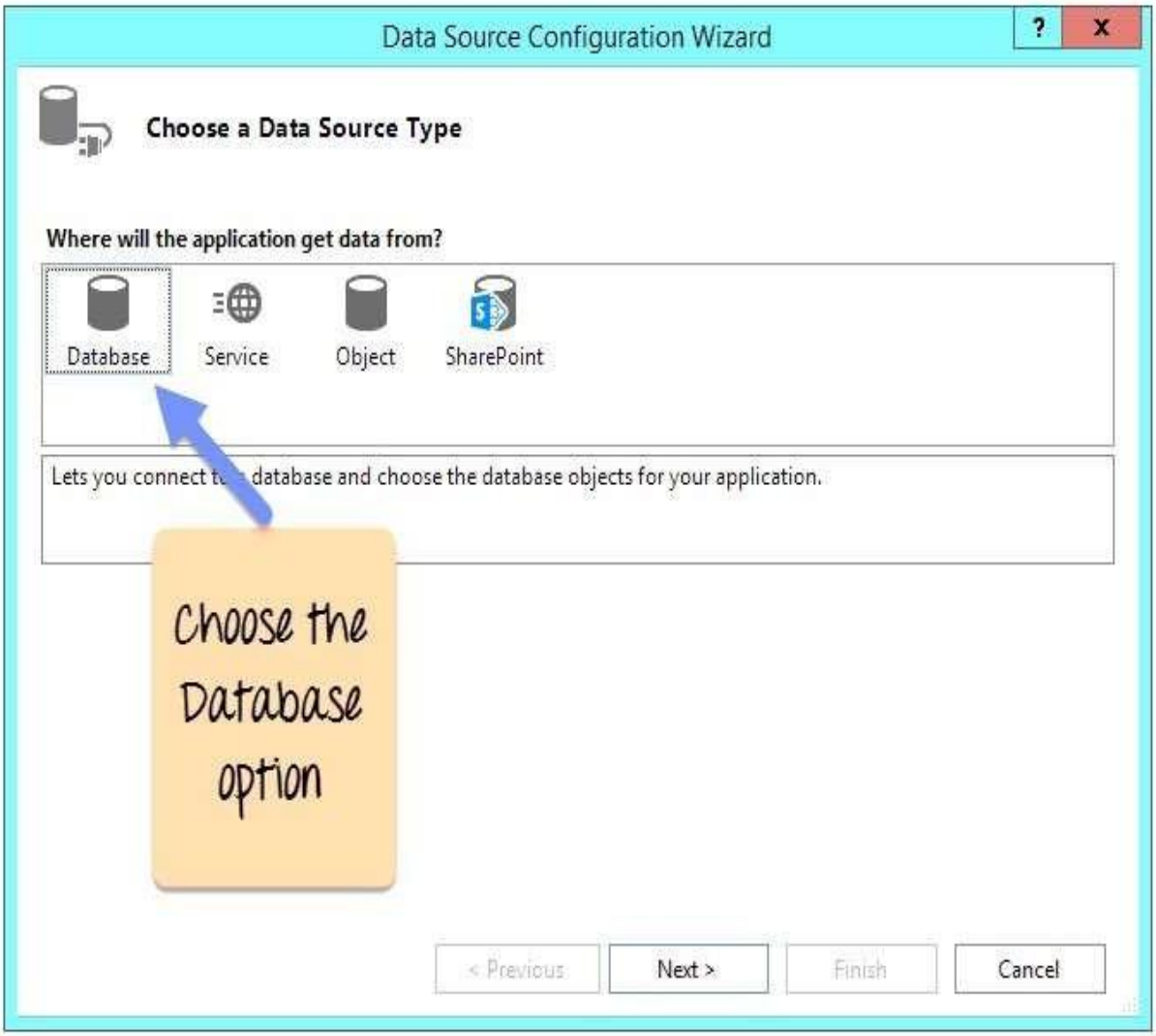
3.5.3 Availability

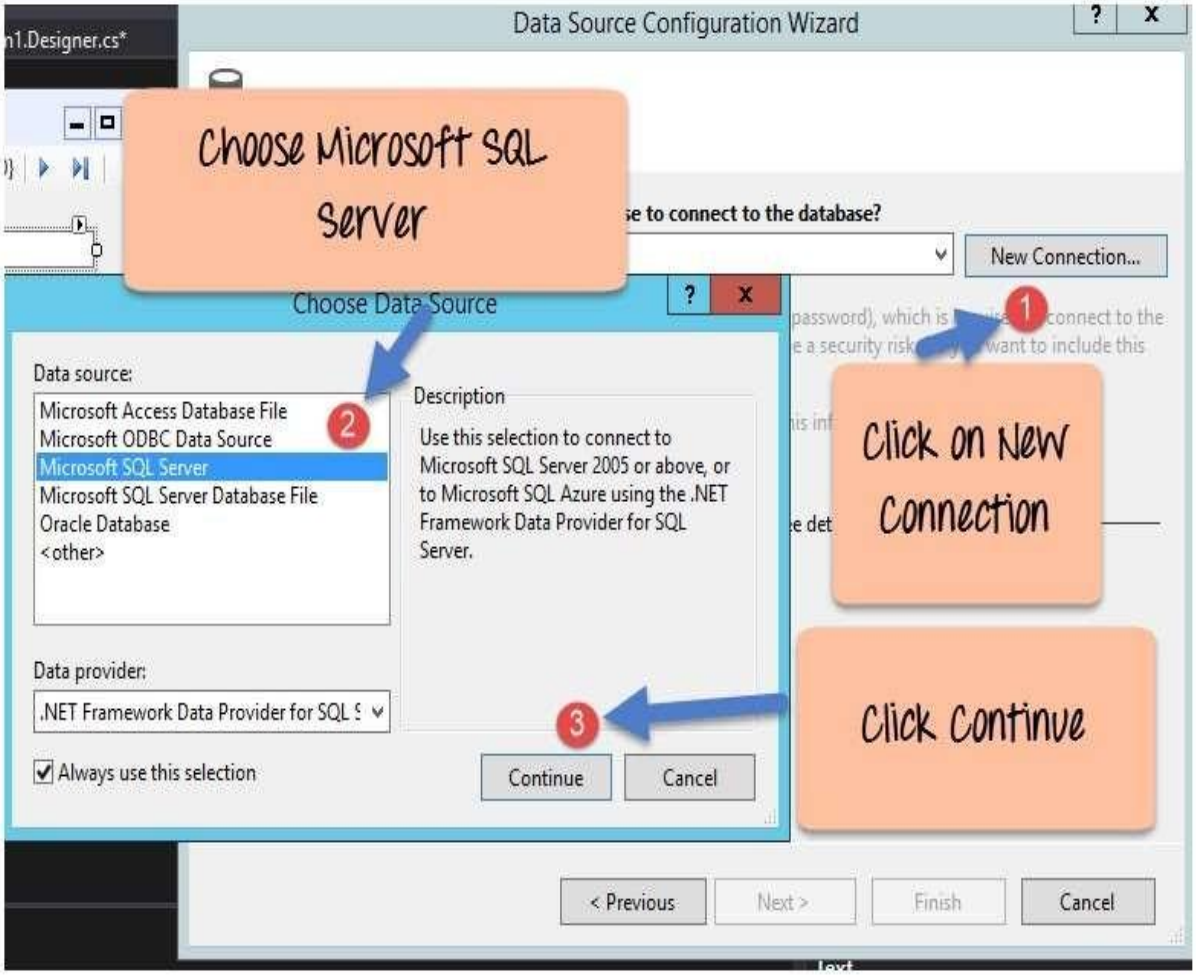
3.5.4 Maintainability

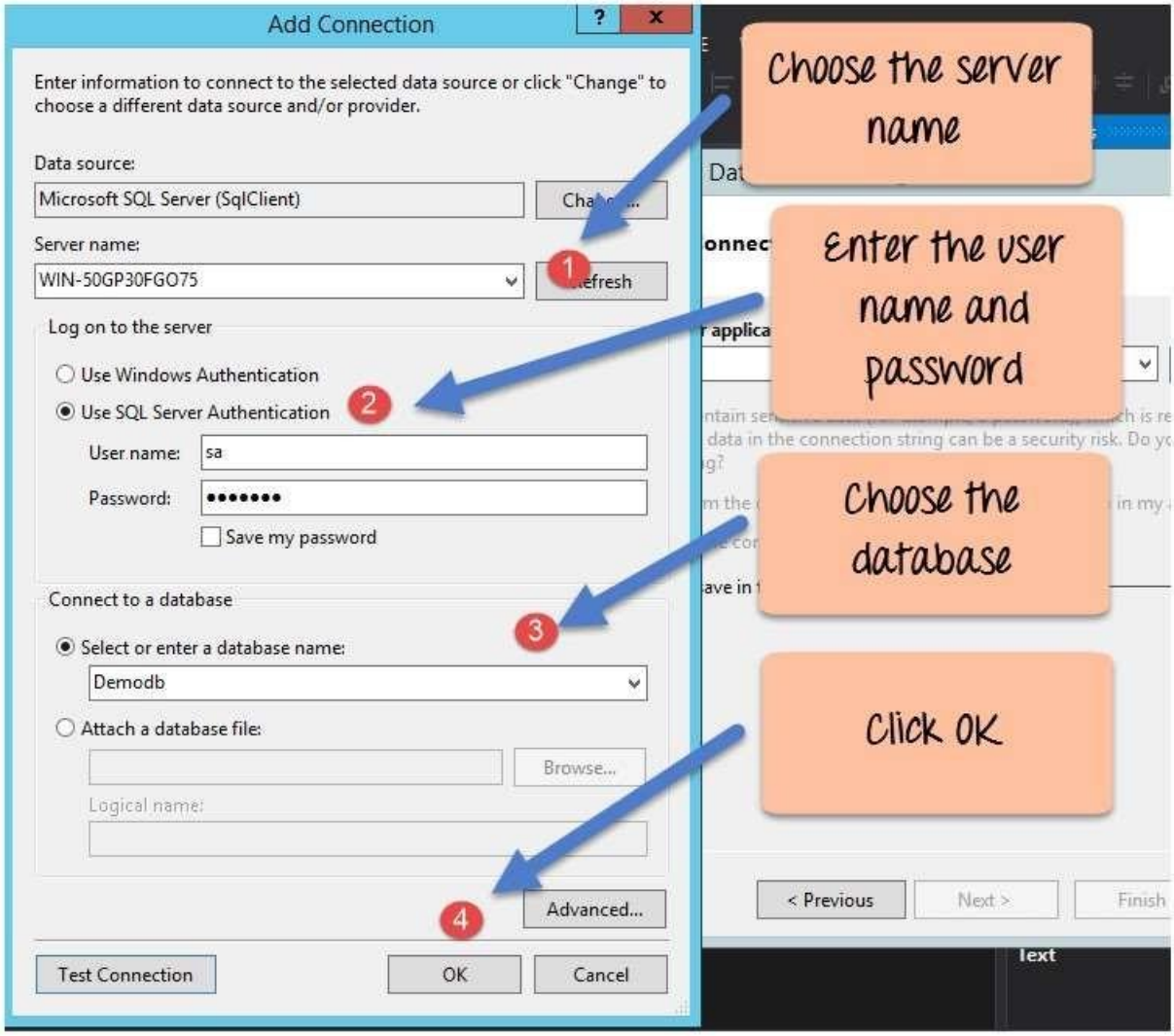
3.6 Inverse Requirements

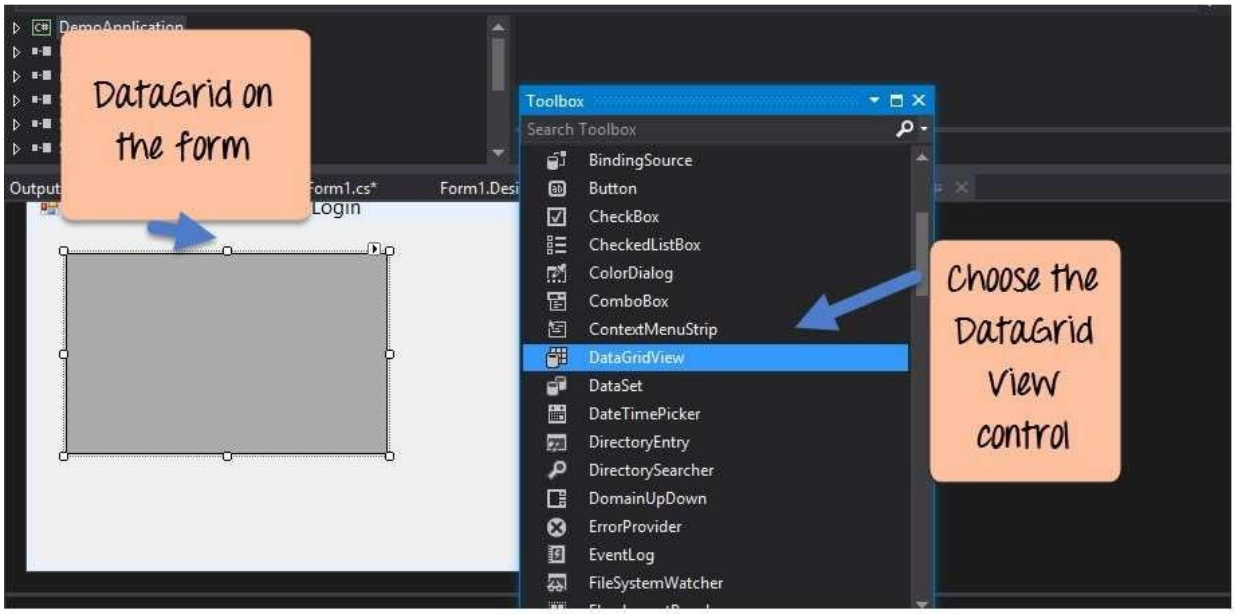
It is very significant to keep up capable software to manage information of a Desktop Application. It's not the dynamic application user can only view the information. He can only access the data that are allowed to user.

3.7 Logical Database Requirements




Data Source Configuration Wizard
Choose Microsoft SQL Server
New Connection...
Choose Data Source
Data source:
Microsoft Access Database File
Microsoft ODBC Data Source
Microsoft SQL Server
Microsoft SQL Server Database File
Oracle Database
<other>
Description
Use this selection to connect to Microsoft SQL Server 2005 or above, or to Microsoft SQL Azure using the .NET Framework Data Provider for SQL Server.
Click on New Connection
Data provider:
.NET Framework Data Provider for SQL S
Always use this selection
Continue
Cancel
Click Continue
< Previous
Next >
Finish
Cancel


Add Connection
Enter information to connect to the selected data source or click "Change" to choose a different data source and/or provider.
Data source:
Microsoft SQL Server (SqlClient)
Server name:
WIN-50GP30FGO75
Log on to the server
Use Windows Authentication
Use SQL Server Authentication
User name:
sa
Password:
Save my password
Connect to a database
Select or enter a database name:
Demodb
Attach a database file:
Browse...
Logical name:
Advanced...
Test Connection
OK
Cancel
Choose the server name
Enter the user name and password
Choose the database
Click OK
< Previous
Next >
Finish



3.8 Design Constraints

3.8.1 Standards Compliance

Parent Component:

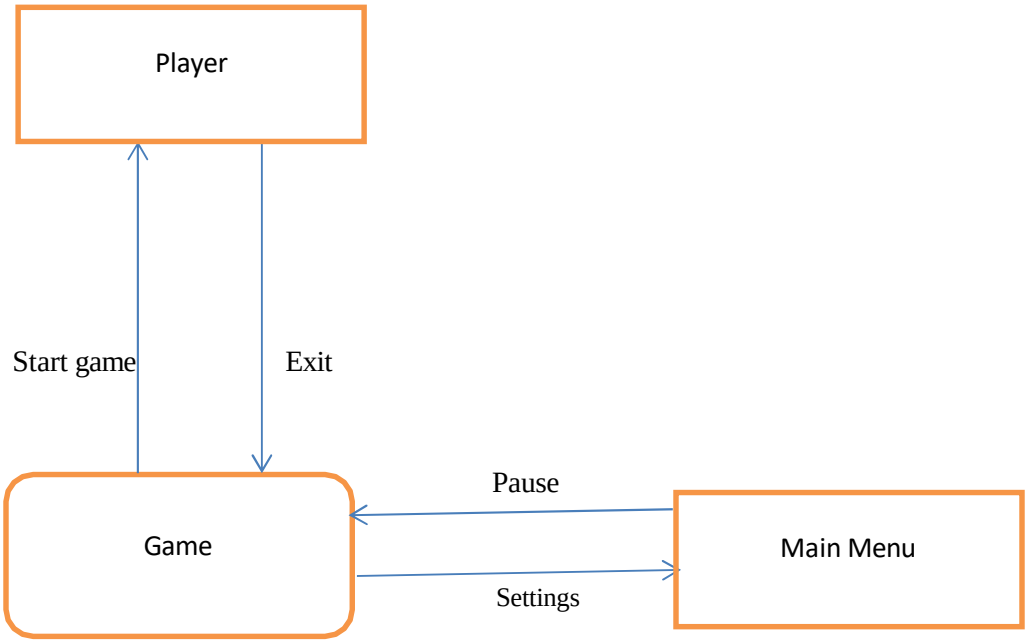
This system is focus on the Windows based application.

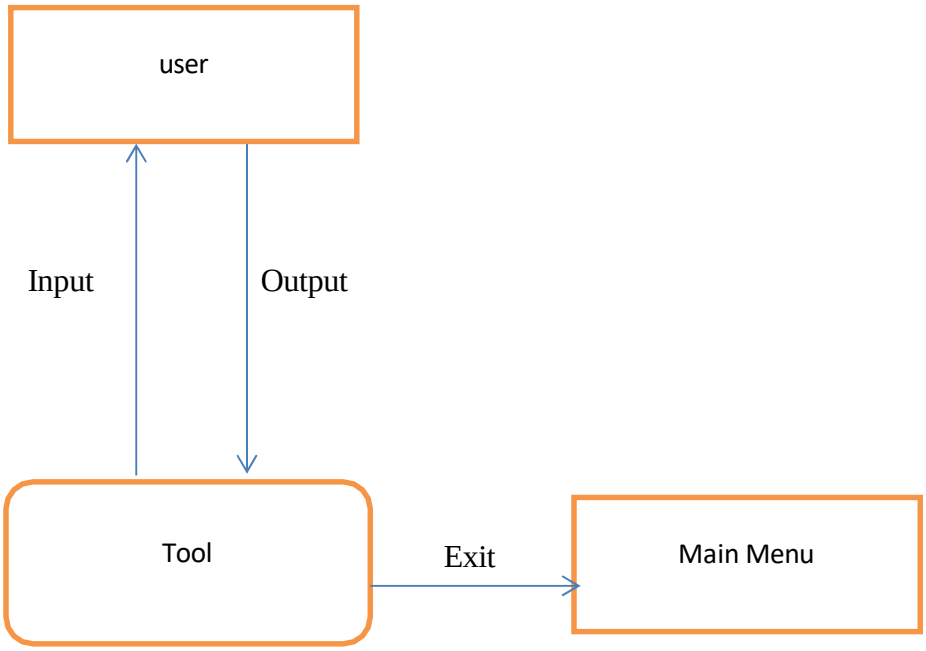
Software Language

C# language is used to develop this application.

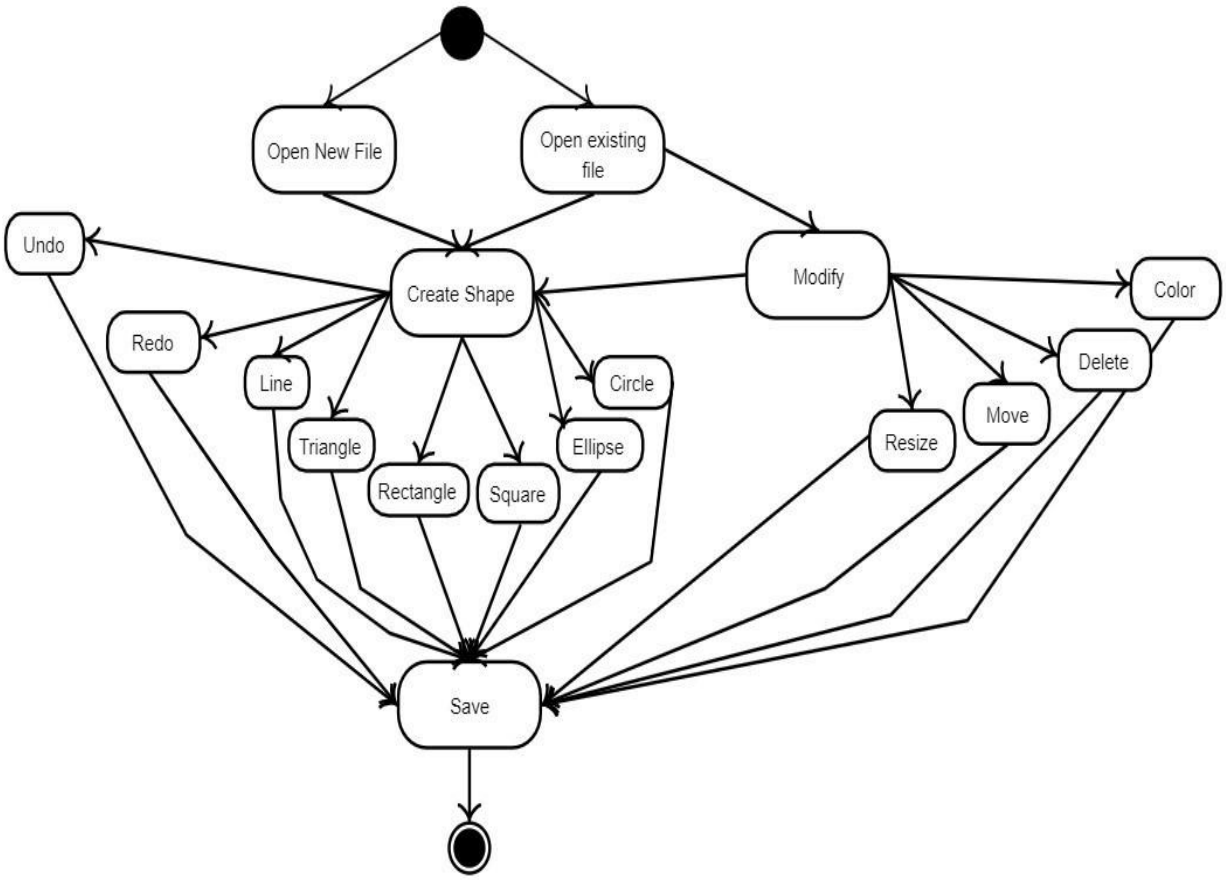
4. Analysis Models

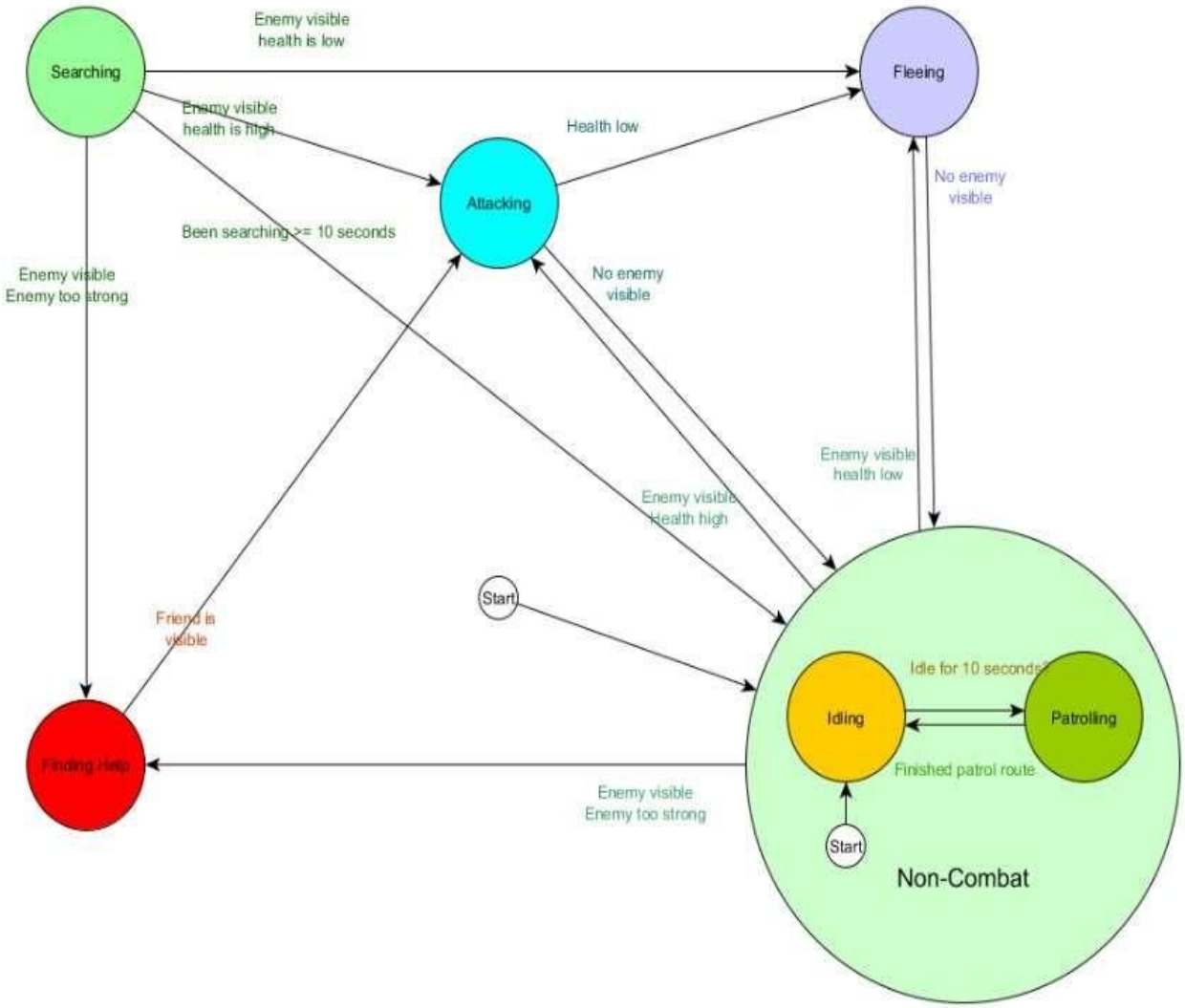
4.1 Data Flow Diagrams (DFD)




user
Input
Output
Tool
Exit
Main Menu

4.3 State-Transition Diagrams (STD)




Searching
Enemy visible
health is low
Fleeing
Enemy visible
health is high
Health low
Attacking
No enemy
visible
Been searching >= 10 seconds
Enemy visible
Enemy too strong
No enemy
visible
Enemy visible
health low
Enemy visible
Health high
Start
Friend is
visible
Idle for 10 seconds?
Idling
Patrolling
Finished patrol route
Finding Help
Enemy visible
Enemy too strong
Start
Non-Combat

5. Supporting Information

Microsoft Windows 10 professional

C#

My SQL

Visual Studio

Microsoft Word 2019

Microsoft Visio 2017

Microsoft Paint

Appendix A

Glossary

Actor

It represents a role, an external entity who interact with our system.

Use case

Graphical Represent of interaction among system and actor.

Scenario

The time of actual input and expected output.

Use case diagram

Represent use the services and functionality by the system by the actor.

Use case Realization

How the use case realizes in the design table model.

Activity diagram

Graphical represent of the process flow of use case etc.

Sequence diagram

Represents the interaction of an object to perform a job.

Collaboration diagram

Alternative representation of sequence diagram.

Data model

Represent the persistence object model.

Appendix B

Web References:

[General]

http://www.yourhtmlsource.com/starthere/whatishtml.html

http://www.w3.org/standards/webdesign/htmlcss

https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Introduction

https://en.wikipedia.org/wiki/PHP

https://dev.mysql.com/doc/refman/5.1/en/what-is-mysql.html

http://www.computerhope.com/jargon/d/dreamweaver.htm

https://en.wikipedia.org/wiki/Firefox

[Search]

http://www.google.com

http://www.37.com

http://www.yahoo.com

http://www.msn.com

http://www.mamma.com

http://www.altavista.com

http://www.ask.com

http://www.registration.com